Power Factor Control

Class Notes

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Vtg / Reactive power / power factor control methods :

Receiving End methods :

-> Shunt Capacitor

-> Shunt Reactor

-> Synchronous Phase Modifier (SPM)

-> Series Capacitor

-> On Load Tap changing X'mer (OLTC)

$$\cos\theta = \frac{P_{RL}}{P_R}$$

Power factor is a degree at which a given R-L or R-C load avg active power consumption compared to pure resistive load.

Advantage of higher pf :

$$I = \frac{P}{V\cos\phi}$$

$\cos\phi\uparrow$, $I\downarrow$, $P_L\downarrow$, $\eta\uparrow$, Heating $\downarrow$, cooling cost $\downarrow$, $V_{reg}\downarrow$

$\sin\phi\downarrow$, $Q\downarrow$, Q burden on Generator $\downarrow$, cost of Energy $\downarrow$

Shunt Capacitor Compensation :

By using shunt capacitor :

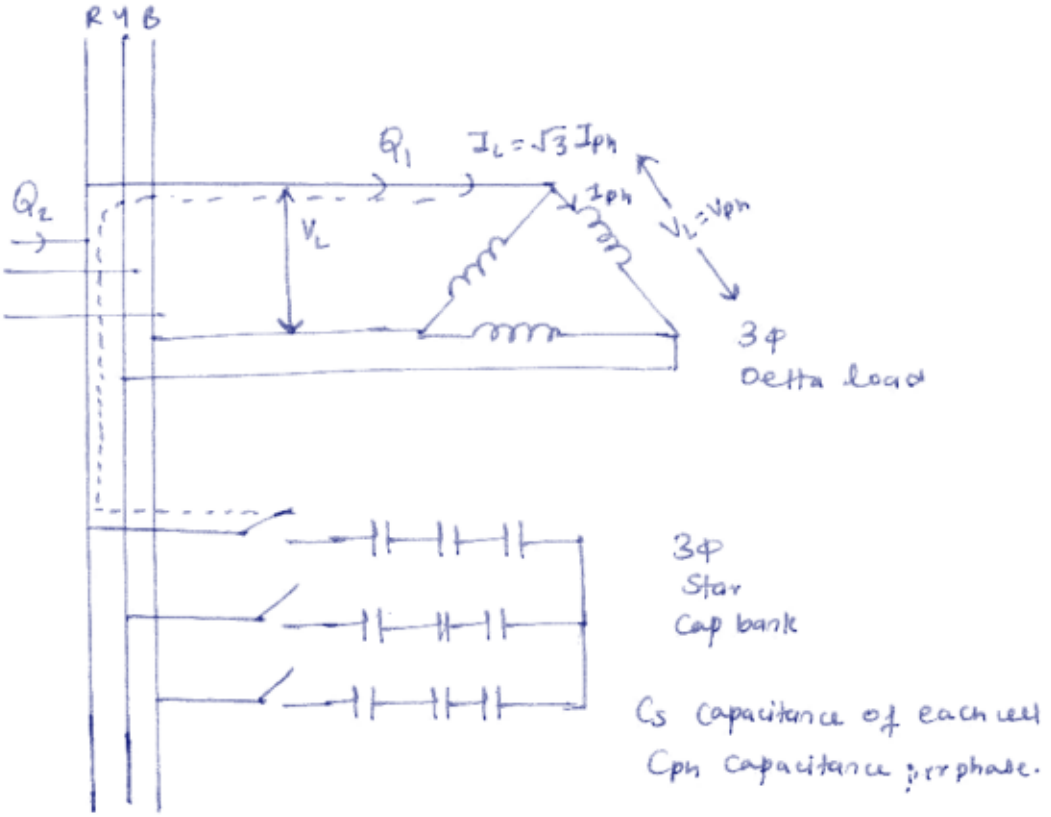
(1) Power factor of the system is improved.

(2) Vtg Reg is changed from +ve to zero or Negative.

(3) Power T/F capability of the line is improved.

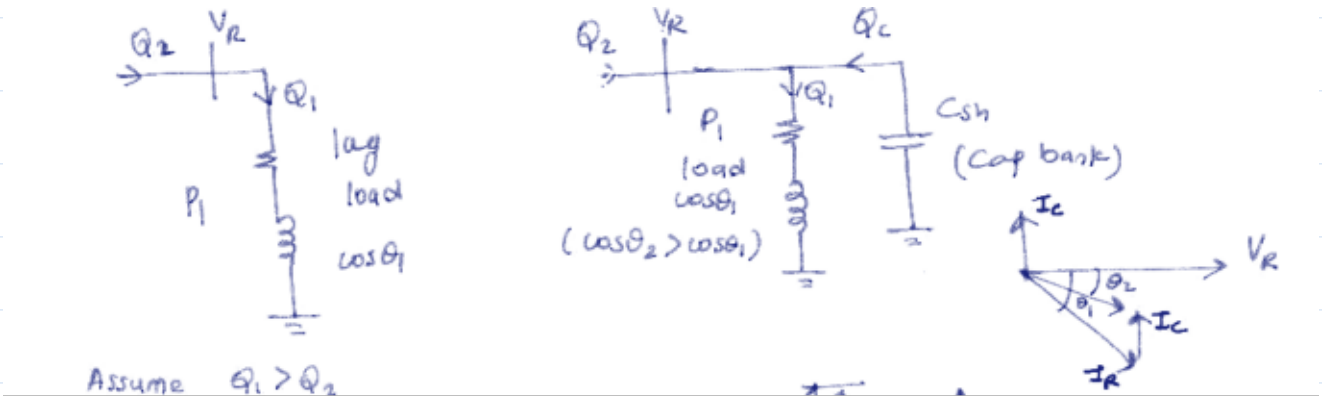
(4) Load Angle (δ) is reduced so that system stability is improved.

Because of the above advantages capacitor banks 5, 7.5, 10 MVAR rating are installing in the 33, 66 and 132 kV S/S and they are kept in service during peak load period.



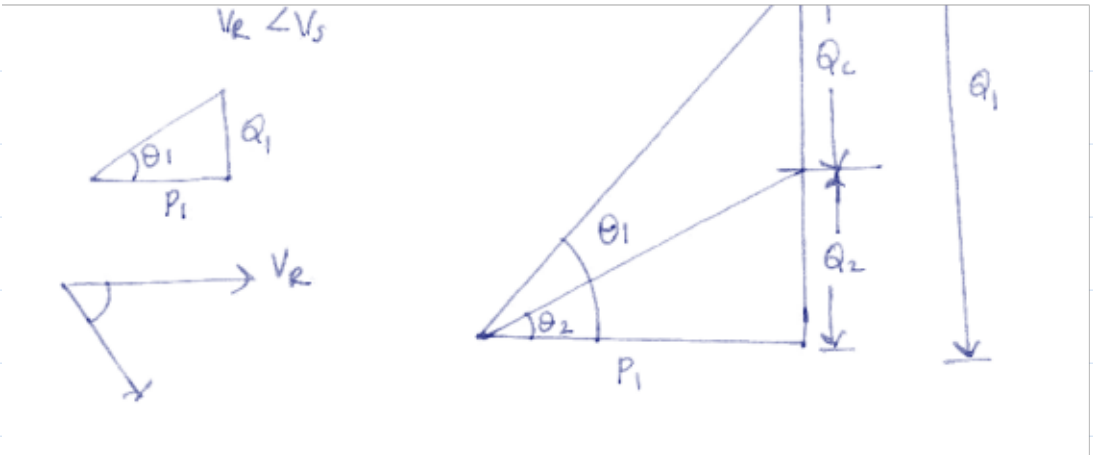
Three-phase delta load (IL = sqrt3 Iph, Q1) supplied from R, Y, B bus with Q2; 3-phase star capacitor bank (series cells); Cs = capacitance of each cell, Cph = capacitance per phase

Without compensator : with C_{sh}



Without compensator: supply Q2, VR, lag load P1, Q1 (cos theta1). With Csh: capacitor bank supplies Qc, load P1, Q1, cos theta2 > cos theta1; phasors VR, IR, Ic, theta1, theta2

Assume $Q_1 > Q_2$, $V_R < V_S$



Power triangles: P1, Q1, theta1; with compensation Qc, Q2, theta2

3-ϕ Rating of C_{sh} : $[Q_{c3\phi}]$

$$Q_1 = Q_2 + Q_c, \quad Q_c = (Q_1 - Q_2) = Q_{c3\phi} = P_1(\tan\theta_1) - P_1(\tan\theta_2)$$

$$\boxed{Q_c = P_1(\tan\theta_1 - \tan\theta_2)} \quad \text{(GATE)}$$

P_1 = 3-ϕ Avg. Active power (I/P to load) = $\frac{P_o}{\eta}$ = O/P / Efficiency = (P_{max}) x Plf (plant load factor)

$$P_1 = \left/ \frac{3V_L^2}{Z_1} \right/ \cos(\theta_1) \cdots \Delta \text{ load}; \quad = \left/ \frac{V_L^2}{Z_1} \right/ \cos(\theta_1) \cdots Y \text{ load}$$

Delta load : $P_{3\phi} = \sqrt{3}V_L I_L \cos\phi_1 = \sqrt{3}V_L(\sqrt{3}I_{ph}\cos\phi_1) = 3\left/ \frac{V_L^2}{Z} \right/ \cos\phi_1$;

$Q_{3\phi} = 3\frac{V_L^2}{Z}\sin\phi_1$

Y load : $P_{3\phi} = \sqrt{3} V_L I_L \cos\phi_1 = \sqrt{3} V_L \frac{V_{ph}}{Z} \cos\phi_1 = \sqrt{3} V_L \frac{V_L}{\sqrt{3}Z} \cos\phi_1 = \left/ \frac{V_L^2}{Z} \right/ \cos\phi_1$;

$Q_{3\phi} = \left/ \frac{V_L^2}{Z} \right/ \sin\phi_1$

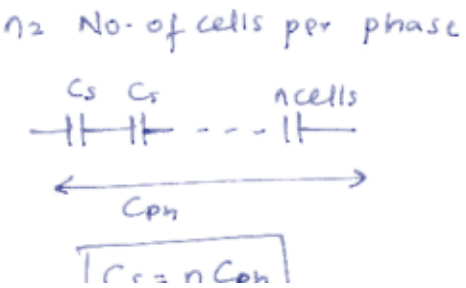
Capacitance per phase [C_{ph}] :

$$Q_{cph} = \frac{Q_{c3\phi}}{3} = 2\pi f V_{ph}^2 C_{ph}$$

$$\boxed{C_{ph} = \frac{Q_{c3\phi}}{3(2\pi f V_{ph}^2)} = \frac{P_1(\tan\theta_1 - \tan\theta_2)}{3(2\pi f V_{ph}^2)}} \quad \text{(GATE)}$$

where $V_{ph} = V_L$ for Δ cap bank ; $V_{ph} = \frac{V_L}{\sqrt{3}}$ for Y cap bank

Capacitance of each cell (C_s) ; n = No. of cells per phase



n cells in series, each Cs, giving Cph per phase

$$\boxed{C_s = n C_{ph}}$$

Shunt Reactor Compensation :

By using shunt reactor :

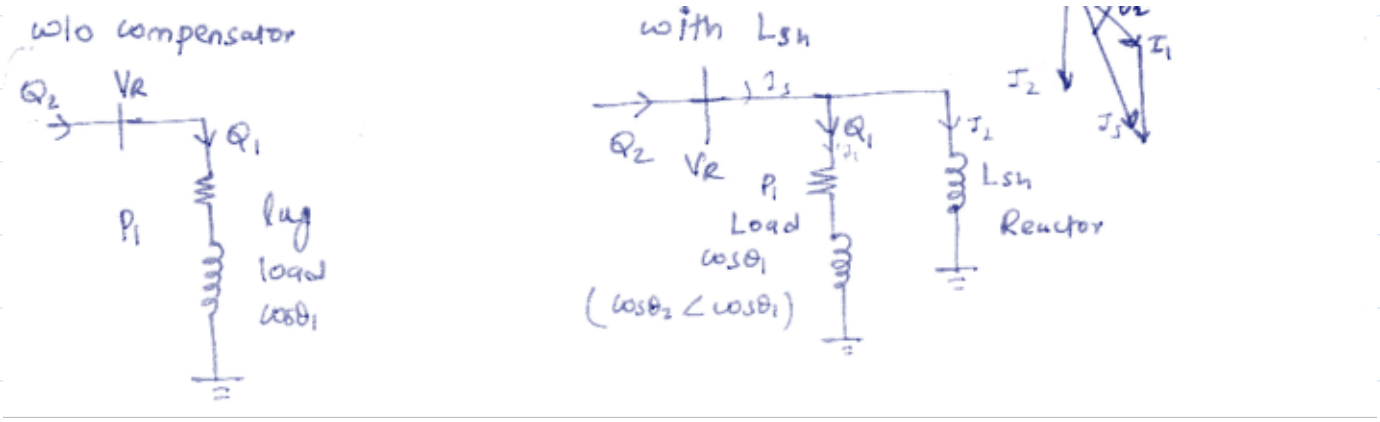
(1) Ferranti effect is compensated.

(2) Vtg Reg changes from Negative to zero or positive.

(3) Power T/F capability is reduced.

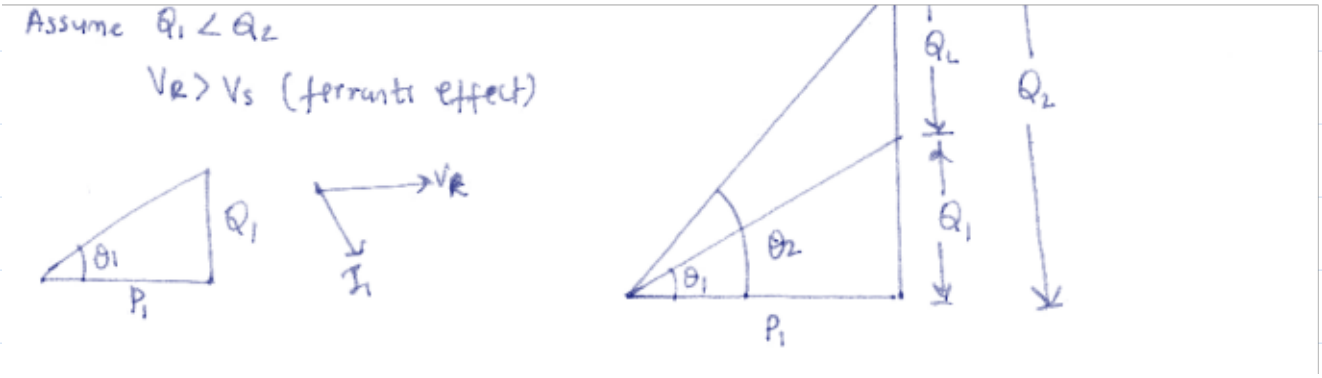
(4) System stability is reduced because load angle δ is increased.

(5) Shunt Reactors are practically used in 400, 765 kV S/S of 80, 125 MVAR to protect the S/S equipment and line from over vtg.



Without compensator: supply Q2, VR, lag load P1, Q1, cos theta1. With Lsh: reactor takes Q_L, load P1, cos theta2 < cos theta1; currents I1, I2, I3; VR, theta2

Assume $Q_1 < Q_2$; $V_R > V_S$ (Ferranti effect)



Power triangles: P1, Q1, theta1; QL, Q1, Q2, theta2

3-ϕ Rating of L_{sh} : $[Q_{L3\phi}]$

$$Q_2 = Q_1 + Q_L, \qquad Q_L = (Q_2 - Q_1) = Q_{L3\phi} = P_1\tan\theta_2 - P_1\tan\theta_1$$

$$\boxed{Q_L = P_1[\tan\theta_2 - \tan\theta_1] = 3Q_{Lph}}$$

Inductance per phase $[L_{ph}]$:

$$Q_{Lph} = \frac{V_{ph}^2}{2\pi f L_{ph}} = \frac{Q_L}{3}$$

$$\boxed{L_{ph} = \frac{3V_{ph}^2}{2\pi f Q_{Lph}}}$$

Relation between Q, f, V (Imp for GATE)

C_{sh} : $Q = 2\pi f C V^2$

$$\boxed{Q \propto V^2 f}$$

$$\frac{Q_2}{Q_1} = \left(\frac{V_2}{V_1}\right)^2 \left(\frac{f_2}{f_1}\right)$$

L_{sh} : $Q_{Lph} = \frac{V_{ph}^2}{2\pi f L_{ph}}$

$$\boxed{Q \propto \frac{V^2}{f}}$$

$$\frac{Q_2}{Q_1} = \left(\frac{V_2}{V_1}\right)^2 \times \left(\frac{f_1}{f_2}\right)$$

SPM (Synchronous Phase Modifier)

* SPM is basically a synchronous motor running under no load condition (load angle δ is 5°) while controlling the excitation, i.e. Vtg of the motor is controlled so that reactive power is controlled.

* An overexcited SPM works similar to the shunt capacitor, absorbing the leading VAR and delivering the lagging VARs.

* An underexcited SPM absorbs lagging VAR and delivers leading VAR, working similar to the shunt reactor.

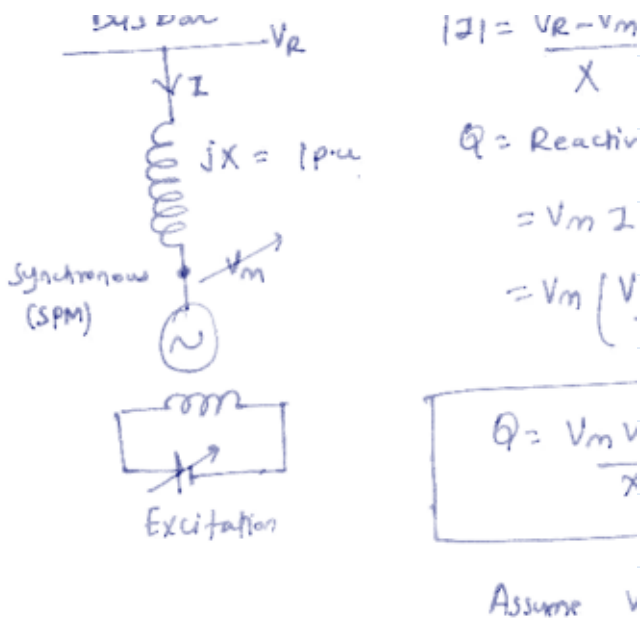
Disadvantages of SPM :

(1) Dynamic device, so it consumes both Active and Reactive power.

(2) Starting mechanism is required to bring the synchronous motor up to synchronous speed.

(3) Requires more maintenance and cooling medium for reducing the heat developed due to rotations.

(4) To overcome the above disadvantages, in modern S/S power electronics based converters called static compensator (STATCOM) are used, which do not have rotating parts.



Busbar VR, current I, reactance jX = 1 p.u., synchronous motor (SPM) with voltage Vm and variable excitation

$$|I| = \frac{V_R - V_m}{X}$$

Q = Reactive power drawn by motor $= V_m I = V_m \left(\frac{V_R - V_m}{X} \right)$

$$\boxed{Q = \frac{V_m V_R}{X} \left(1 - \frac{V_m}{V_R} \right)}$$

Assume $V_R = 1$ p.u. , $X = 1$ p.u.

$$Q = V_m (1 - V_m)$$

(1) Normal Excitation $[V_m = V_R = 1 \text{ p.u.}] \rightarrow Q = 0$

(2) Over excited $[V_m > V_R]$; Assume $V_m = 1.2 > 1$; $Q = 1.2(1 - 1.2) = -0.24$; SPM absorbs leading Var, hence working similar to C_{sh}

(3) Underexcited $[V_m < V_R]$; $V_m = 0.9$ p.u. < 1 p.u. ; $Q = 0.9(1 - 0.9) = 0.09$ p.u. ; SPM absorbs lag Vars, hence working similar to L_{sh}.

(a) Overexcited SPM [C_{sh}]

$(\cos\theta_2 > \cos\theta_1)$

Rating of SPM = $S = (P_s - jQ_c)$

p.f of SPM = $\cos\left[\tan^{-1}\frac{Q_c}{P_s}\right]$

$$Q_c = P_1\tan\theta_1 - (P_1 + P_s)\tan\theta_2$$

Supply Q2, bus VR, lag load P1, Q1 cos theta1, SPM supplies Qc; cos theta2 > cos theta1

Rating of SPM = $S = (P_s - jQ_c)$; p.f. of SPM = $\cos\left[\tan^{-1}\frac{Q_c}{P_s}\right]$

$$Q_c = P_1\tan\theta_1 - (P_1 + P_s)\tan\theta_2$$

(b) Underexcited SPM [L_{sh}]

$\cos\theta_2 < \cos\theta_1$

Rating of SPM = $S = P_s + jQ_L$

p.f of SPM = $\cos\left[\tan^{-1}\frac{Q_L}{P_s}\right]$

$$Q_L = (P_1 + P_s)\tan\theta_2 - P_1(\tan\theta_1)$$

Supply Q2, bus VR, lag load cos theta1, SPM absorbing; cos theta2 < cos theta1

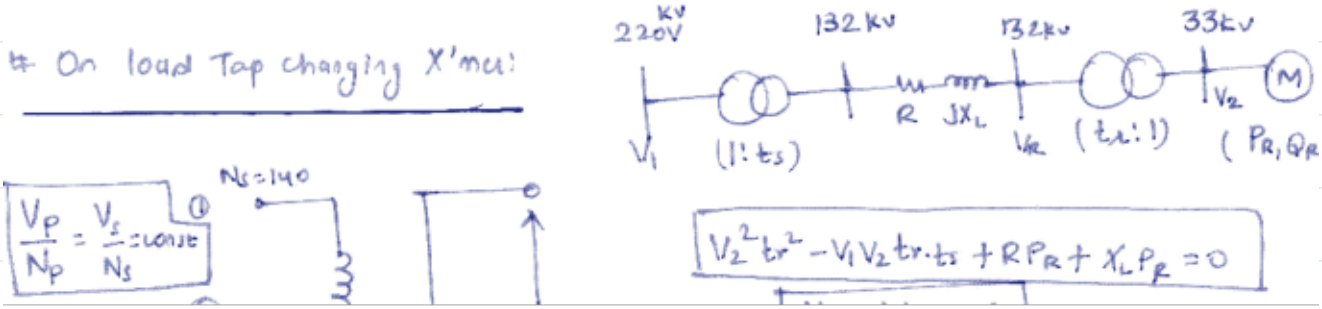
Rating of SPM = $S = P_s + jQ_L$; p.f. of SPM = $\cos\left[\tan^{-1}\frac{Q_L}{P_s}\right]$

$$Q_L = (P_1 + P_s)\tan\theta_2 - P_1(\tan\theta_1)$$

P_s = 3-ϕ Avg Active power I/P to SPM = $\frac{P_o}{\eta}$

On load Tap changing X'mer (OLTC)

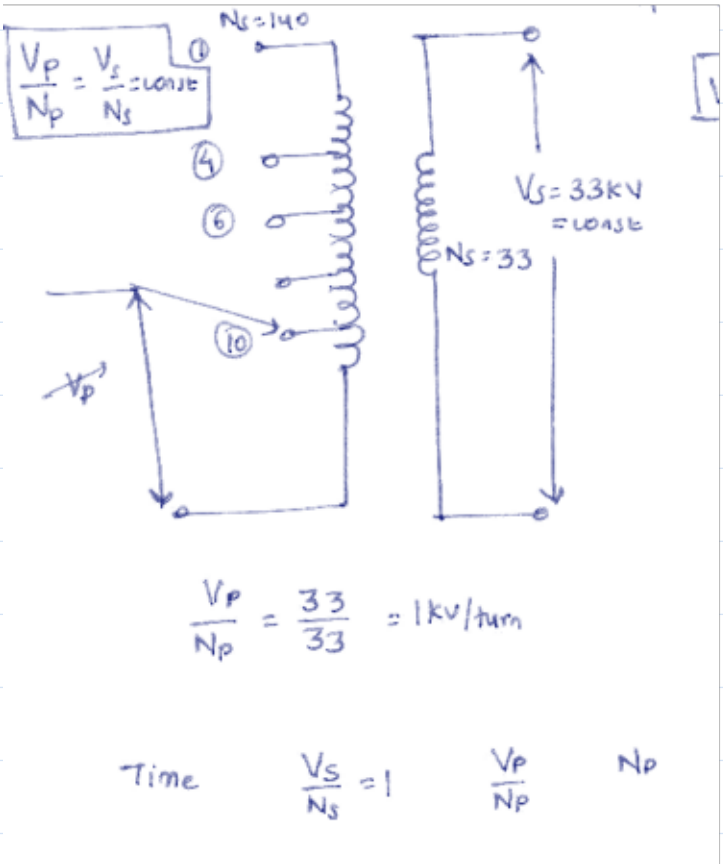
On load Tap changing X'mer:



220 kV bus V1, transformer (1:ts), 132 kV line R + jXL, receiving end VR, transformer (tr:1), 33 kV bus V2, motor load PR, QR

$$V_2^2 tr^2 - V_1 V_2 tr \cdot ts + R P_R + X_L P_R = 0$$

$$V_2 = f(tr, ts)$$



Tapped primary winding Np = 140 (taps 1, 4, 6, 10), secondary Ns = 33, Vs = 33 kV = constant

$$\frac{V_p}{N_p} = \frac{V_s}{N_s} = \text{constant} ; \frac{V_p}{N_p} = \frac{33}{33} = 1 \text{ kV/turn}$$

* OLTC taps are arranged on HV side because current magnitude is less on HV side and hence intensity of arc is less during changing of taps, which will increase the life of X'mer.

Time | Vs/Ns = 1 | Vp/Np | Np | Tap

4 AM | 1 | 140 | 140 | 1

8 AM | 1 | 136 | 136 | 4

10 AM | 1 | 132 | 132 | 6

4 PM | 1 | 120 | 120 | 20

5 PM | 1 | 115

* Make before break technique is used to change the taps from 1 position to other position, so that there is no interruption of supply during changing of Taps.